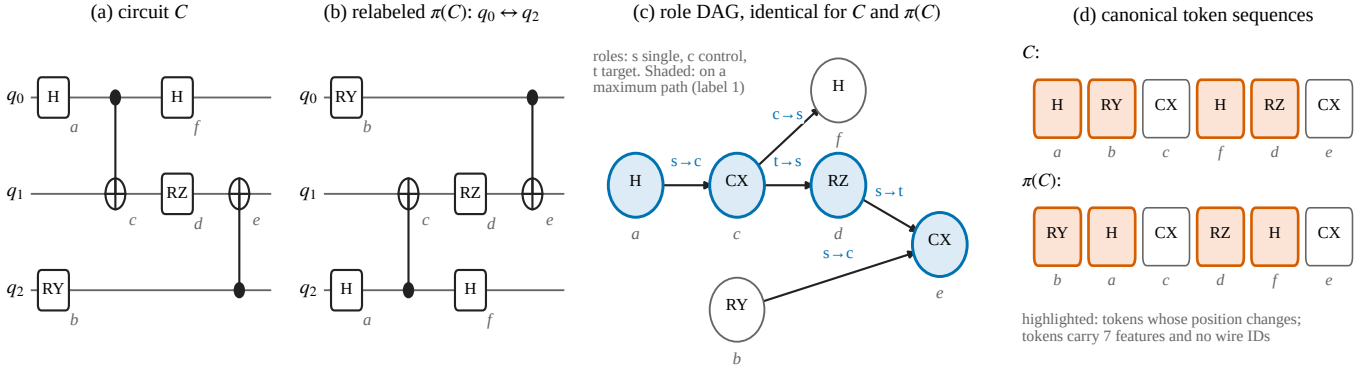
(a) circuit C
(b) relabeled $\pi(C)$: $q_0 \leftrightarrow q_2$
(c) role DAG, identical for C and $\pi(C)$
(d) canonical token sequences
q_0
q_1
q_2
H
RY
RZ
CX
a
b
c
d
e
f
roles: s single, c control, t target. Shaded: on a maximum path (label 1)
c → s
t → s
s → c
s → t
s → c
C:
$\pi(C)$:
highlighted: tokens whose position changes; tokens carry 7 features and no wire IDs